

5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Date

June 1-4, 2026

Venue

Santiago, Chile



Slide 1 / 67 ~42s cum 0:42

CORE

Title

Say (verbatim):

- Good morning, everyone. Thank you for being here. My name is Jorge Mayorga, from Universidad de los Andes.
- I want to open with a question that sounds simple but really is not: in a power grid that is increasingly run by inverters, which frequency estimator should we actually trust?
- We tend to assume the estimator is a solved, interchangeable block. It is not.
- The honest answer to that question is, it depends, and the whole point of this talk is to make 'it depends' precise, measurable, and useful for engineers.

→ *So let me begin with why this is a problem worth solving.*

Key line: There is no universal best estimator; the right choice depends on what you measure.

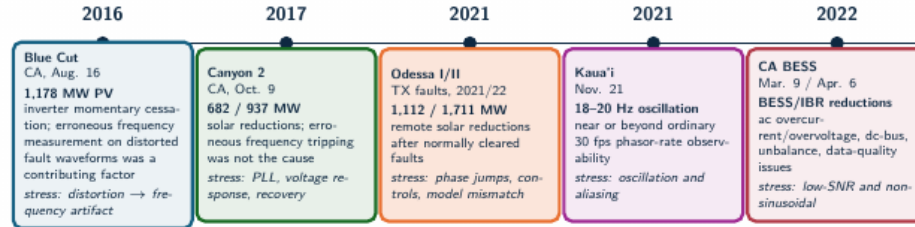
Motivation (section)

Key line: Why this matters.

Motivation

Why frequency estimation in low-inertia IBR grids is hard

Field events define the benchmark stress model



Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC/WECC Blue Cut 2017; NERC/WECC Canyon 2 2018; NERC Odessa 2022; NERC/WECC CA BESS 2023; Dong et al. Kaua'i 2021 analysis

Field events define the stress model

Say (verbatim):

- Let me ground this in reality. Real inverter-driven disturbances look nothing like the clean sine wave we use in textbooks.
- If you look at recorded events, Blue Cut, Odessa, the Kauai oscillation, you see phase jumps, fast rate-of-change of frequency, harmonics, and noise all arriving at the same time.
- So instead of inventing convenient test signals, we let these real field events define the stress cases that our benchmark has to cover.
- That keeps the evaluation honest, because the grid does not care about our assumptions.

→ And here is the catch: the very instrument we use to watch these events also shapes what we conclude about them.

Key line: Field events, not textbook signals, set the bar.

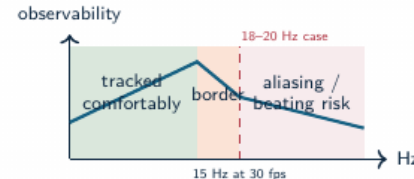
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

PMU streams show consequences

Say (verbatim):

- A standard PMU hands us a frequency value and a rate-of-change of frequency.
- But during a fast inverter event, those numbers show the consequence, not always the cause.
- And crucially, it is the estimator buried inside the PMU that decides what we actually see on the screen.
- Two different estimators, looking at the very same event, can report two different stories.

→ So before we can trust any analysis of an event, we first have to ask which estimator produced those numbers.

Key line: Choosing the estimator is itself a measurement decision.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Research problem

Say (verbatim):

- And that points straight at the gap we are addressing. Today there is no agreed, systematic way to compare these estimators under inverter-specific stress.
- The compliance tests we do have run on clean, well-behaved signals.
- So they tell us a method passes a standard, but they never tell us how it behaves during a real, messy, composite event.
- We are essentially certifying estimators on the easy cases and hoping they survive the hard ones.

→ *Our response to that gap is one clear claim.*

Key line: We need a fair, inverter-relevant way to compare estimators.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Operating claim

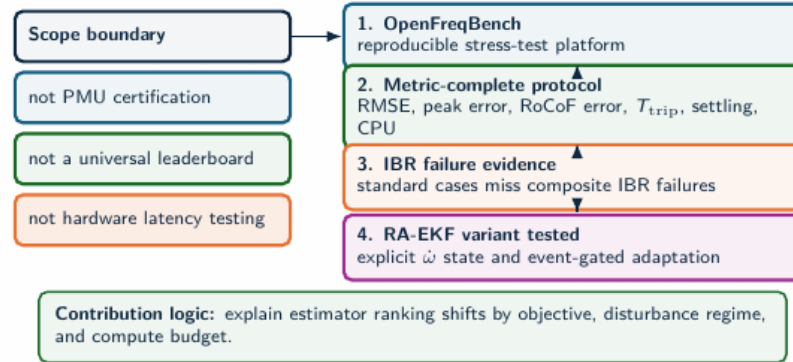
Say (verbatim):

- The central claim of this work is that estimator selection has to be conditional.
- It is not a single ranking. It depends on three things at once: the type of disturbance, the metric you actually care about, and your compute budget.
- Change any one of those, and the best estimator can change with it.
- So our job is not to crown a winner; it is to map out which method wins under which conditions.

→ *And to back that claim with evidence, we built three things.*

Key line: There is no single best estimator, only a best one for a given job.

Technical contributions



Technical contributions

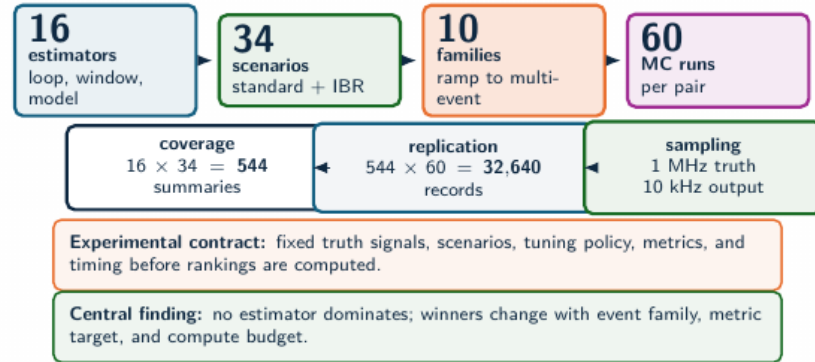
Say (verbatim):

- We make three contributions. First, an open benchmark platform built around known-truth scenarios, so the error is never in doubt.
- Second, a common execution contract, so that all eighteen estimators are tested under exactly identical conditions, with no method getting an unfair advantage.
- And third, a metric vector that keeps accuracy, trip-risk, latency, and computational cost as separate, explicit outputs, instead of collapsing them into one number.

→ *Let me give you a feel for the scale of all this.*

Key line: Open platform, common contract, and multi-metric evidence.

Experiment scale



Source: SGSMA benchmark matrix, Sec. III-IV and Fig. 2 caption

Slide 8 / 67 ~31s cum 4:17

CORE

Experiment scale

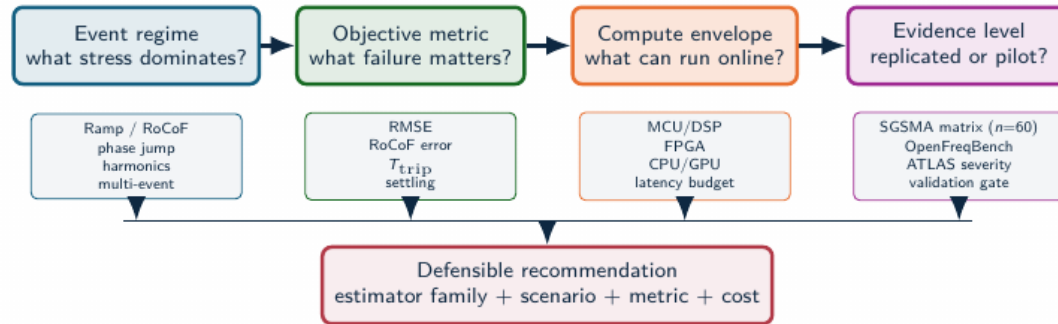
Say (verbatim):

- The scale is deliberately large: eighteen estimators, thirty-three scenarios, across five families of stress, all with Monte-Carlo repetitions.
- On top of that, we built a severity-sweep engine that we call ATLAS, which pushes each stress from mild to extreme.
- And every single result is reproducible from saved artifacts, so anyone can regenerate the figures you will see today.

→ *And the logic for choosing among all of these methods actually fits into a single picture.*

Key line: Large, factorial, and fully reproducible.

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Slide 9 / 67 ~28s cum 4:45

FIG

Estimator selection logic (build-up)

Say (verbatim):

- This diagram is, in a sense, the entire talk in miniature.
 - Selection comes down to three questions: which event class are you facing, which metric matters for your application, and what compute budget do you have.
 - Watch as each question appears, because each one, on its own, can send you to a different estimator.
- *With that logic in hand, let me now show you the platform itself.*

Key line: This one diagram is the whole talk in miniature.

Benchmark and platform (section)

Key line: First, the benchmark and the platform.



Benchmark & platform

What is missing in prior work — and OpenFreqBench

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Benchmark literature gap

Standards
PMU compliance, metrology, calibration

Field data
real events, PMU/DFR streams

Open platforms
OpenPMU, openPDC, event libraries

Single-method studies
detailed but narrow comparisons

Missing layer
many estimators, same truth, same metrics, same policy

Result: controlled synthetic truth isolates estimator behavior under reproducible stress while field events define realism.

Why existing evidence is not enough

Standards define requirements; field events define realism; open PMU ecosystems provide infrastructure. None alone answers which estimator family is defensible under the same disturbance, tuning policy, metric vector, and compute budget.



Estimator-level benchmark: one artifact contract for algorithms, scenarios, seeds, metrics, and plots

Benchmark literature gap

Say (verbatim):

- If you look at the existing literature, most studies compare a handful of estimators on just one or two cases.
- What has been missing is a benchmark that covers the full estimator space, against composite inverter stress, with metrics you can actually reproduce.
- That combination, breadth, realism, and reproducibility, simply did not exist.

→ *So that is exactly what we set out to build.*

Key line: That blank space on the map is the gap we fill.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

Benchmark comparison axes

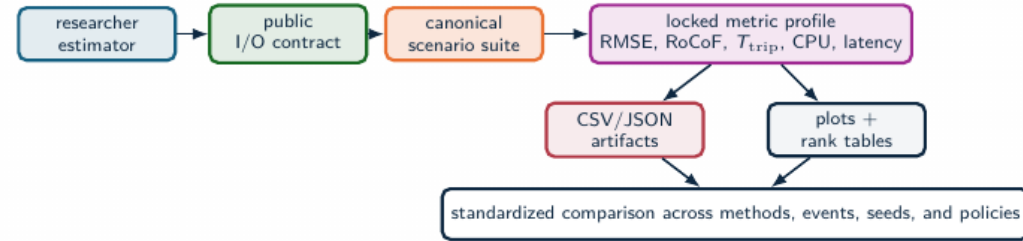
Say (verbatim):

- We compare the methods on four axes at the same time: accuracy, robustness, computational cost, and protection trip exposure.
- And the whole reason we keep them separate is that a method can win brilliantly on one axis and quietly lose on another.
- A single leaderboard would hide exactly the trade-off that an engineer needs to see.

→ *Here is the platform that measures all four of them.*

Key line: One number hides the trade-off; four axes reveal it.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Principle: a submitted estimator becomes a repeatable benchmark entry under the same scenarios, seeds, metrics, and artifacts.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

Slide 13 / 67 ~30s cum 6:13

OpenFreqBench platform

Say (verbatim):

- This is OpenFreqBench. It has four parts.
- A scenario factory that generates waveforms with known true frequency. A uniform interface so every estimator plugs in the same way. A locked metric engine so the scoring never changes between methods. And fully traceable artifacts, every CSV and every figure.
- The design philosophy is simple: known truth in, locked metrics out, and everything traceable.

→ *Let me show you concretely how we keep the comparison fair.*

Key line: Known truth in, locked metrics out, everything traced.

Experimental method (1)

Say (verbatim):

- Fairness is really the heart of this whole project.
- Every estimator sees the exact same input trace, driven by the same random seeds, with the same warm-up handling.
- Nobody gets a cleaner signal or a luckier start. The comparison is fair by construction, not because we trust ourselves to be fair.

→ *And we are just as strict about tuning and about timing.*

Key line: Fairness is enforced by construction, not by trust.



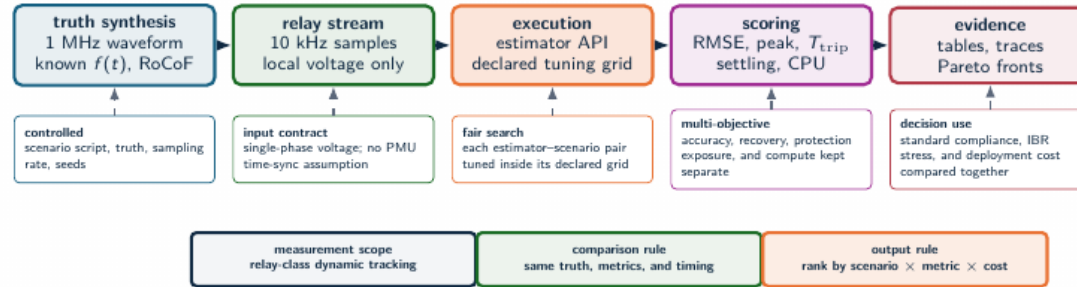
Experimental method

Same truth, same metrics, fair tuning

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Experimental method



Source: Methodology and test framework

Experimental method (2)

Say (verbatim):

- We tune every method under a single fixed policy, we discard the warm-up region before scoring, and we record CPU cost as a genuine output, not as an afterthought.
- So the computational numbers you will see are measured, not assumed.

→ *Put all of that together, and this is the pipeline.*

Key line: Nothing is hand-tuned to flatter one method.

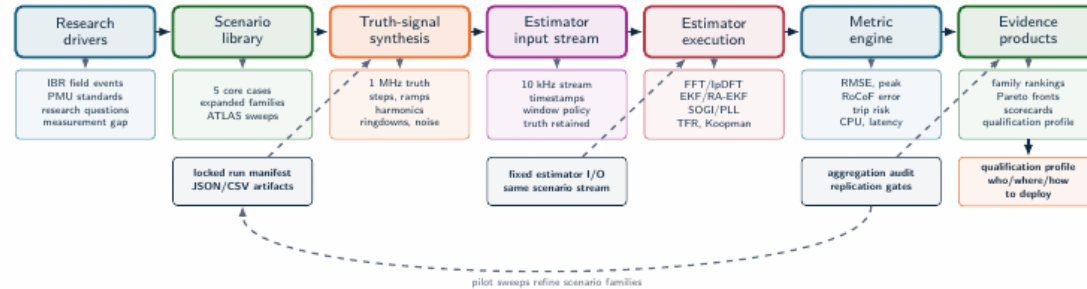
Full benchmark workflow

Say (verbatim):

- The workflow runs in a fixed order: scenario, then Monte-Carlo, then the estimator, then the locked metrics, and finally the artifacts.
- And I want to stress this: every single figure in this talk traces straight back to those artifacts. Nothing here is hand-drawn.

→ *Now, before the scenarios, let me show how someone actually adds a method.*

Full benchmark workflow



The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

Key line: One reproducible pipeline, end to end.

Adding an estimator: one small Python contract

The whole interface a new method must satisfy

```
class MyEstimator(BaseFrequencyEstimator):
    name = "My-Method"

    def _step(self, z, t_s, memory):
        # one input sample z -> one f_hat [Hz]
        ...
        return f_hat

    def structural_latency_samples(self):
        return self.window_len # declared delay

    def tuning_grid(cls):
        return [{...}, {...}] # fair-tuning space
```

Why this is enough

- A method only implements `_step`: *one sample in, one frequency out*. No I/O, no plotting, no metric code.
- `structural_latency_samples` forces every author to **declare** the inherent delay (window/group delay), so latency is comparable, not hidden.
- `tuning_grid` exposes the parameter space the harness sweeps under a single fixed policy — nobody hand-tunes per scenario.

The contract is deliberately tiny: anything an author writes beyond `_step` is logic, never evaluation. That is what makes 18 very different methods comparable.

Adding an estimator: one small Python contract

Say (verbatim):

- This is the entire interface a new estimator has to satisfy. A class, a name, and one method called underscore-step: one input sample in, one frequency estimate out.
- Two optional hooks make the comparison fair. `structural-latency-samples` forces every author to declare their inherent delay, so latency is comparable and never hidden. And `tuning-grid` exposes the parameter space, which the harness sweeps under a single fixed policy, so nobody hand-tunes per scenario.
- Everything an author writes beyond that is algorithm logic, never evaluation code.

→ And the performance numbers are not something the author reports — they are measured for them.

Key line: The contract is tiny on purpose: that is what makes eighteen very different methods comparable.

Performance is measured inside the contract, not by the author

The public step() wrapper instruments every call

```
def step(self, z, t_s, memory):
    t0 = time.process_time()
    y = self._step(z, t_s, mem) # author code
    dt = time.process_time() - t0

    self._times_us.append(dt*1e6) # CPU
    if not isfinite(y):
        self._invalid += 1 # NaN/Inf guard
    mem.observe() # memory proxy
    return y
```

What the harness records per run, for free

- **M13 CPU** — mean process_time per sample (μ s), wrapped around the author's code so it cannot be gamed.
- **M20 jitter** — standard deviation of per-step time.
- **invalid-output rate** — fraction of NaN/Inf returns (divergence detector).
- **startup-valid sample** — when the first usable output appears.
- **memory proxy** — peak/mean bytes via MemoryStore.

Because the timing and validity instrumentation lives in the *wrapper*, every estimator is measured identically — the same anti-cheat path for a 7 μ s ZCD and a 15,000 μ s Koopman.

Performance is measured inside the contract

Say (verbatim):

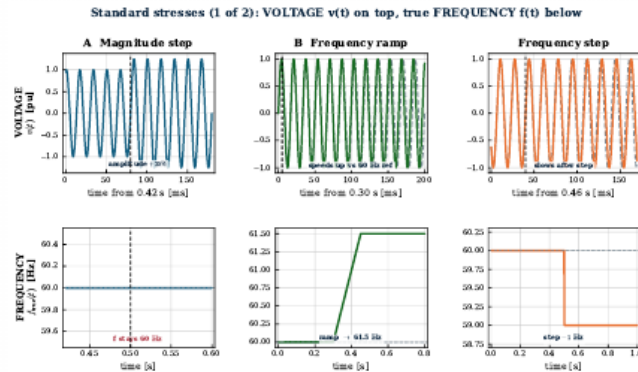
- Here is the trick that keeps it honest. The public step method wraps the author's code: it timestamps with process-time before and after the call, so CPU cost is measured around their logic and cannot be gamed.
- In the same wrapper we catch NaN or infinite outputs, which is our divergence detector, and we sample a memory proxy.
- So for free, every run records CPU per sample, timing jitter, invalid-output rate, the first valid sample, and memory — identically for a seven-microsecond ZCD and a fifteen-thousand-microsecond Koopman.

→ Now let me introduce the scenarios themselves.

Key line: Timing and validity live in the wrapper, so every estimator is measured on the same anti-cheat path.

Standard stresses (1 of 2): magnitude, ramp, frequency step

CORE SCENARIO SUITE



SG: Top row is the voltage waveform $v(t)$; bottom row is the true frequency $f(t)$.

Standard stresses (1 of 2): magnitude, ramp, step

Say (verbatim):

- We start with the standard stresses. On this slide the top row is the voltage waveform, and the bottom row is the true frequency, for three cases: a magnitude step, a frequency ramp, and a frequency step.
- Because we generate the waveform ourselves, we know that true frequency at every single sample, so the error is exact.
- And notice the magnitude step on the left: the voltage clearly changes, but the frequency below it stays flat at sixty hertz.

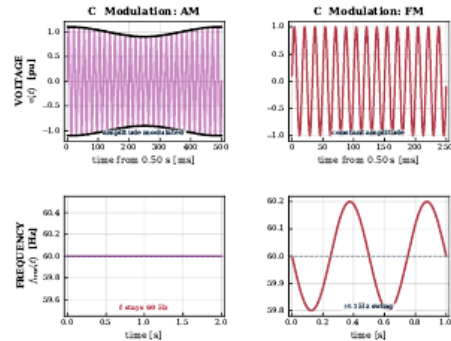
→ *Modulation gives us the other two standard cases.*

Key line: Top row is voltage; bottom row is the true frequency we score against.

Standard stresses (2 of 2): modulation (AM and FM)

CORE SCENARIO SUITE

Standard stresses (2 of 2): VOLTAGE $v(t)$ on top, true FREQUENCY $f(t)$ below



SG:

Voltage on top, true frequency below: AM leaves $f(t)$ flat at 60 Hz, while only



Slide 20 / 67 ~25s cum 9:44

FIG Standard stresses (2 of 2): modulation AM and FM

Say (verbatim):

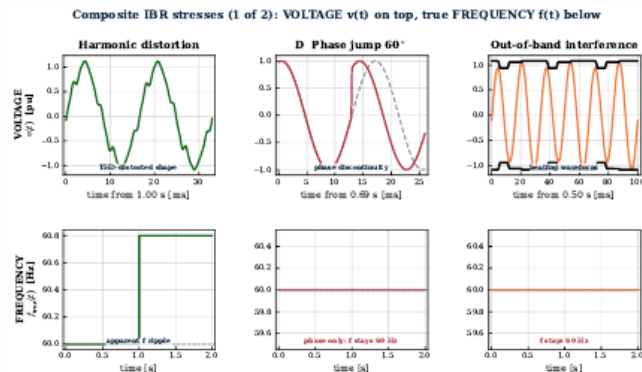
- Here are the two modulation cases, again voltage on top and true frequency below.
- Amplitude modulation, on the left, leaves the frequency perfectly flat at sixty hertz. Only frequency modulation, on the right, actually moves it.
- So an estimator can be disturbed by a signal whose true frequency never changed at all.

→ Now let us make the scenarios composite and realistic.

Key line: AM leaves the frequency flat; only FM truly moves it.

Composite IBR stresses (1 of 2): harmonics, phase jump, out-of-band

IBR STRESS SUITE



Composite IBR stresses (1 of 2): harmonics, jump, OOB

Say (verbatim):

- Now we stack the physics. Top row voltage, bottom row true frequency, for three composite cases: harmonic distortion, a phase jump, and out-of-band interference.
- These badly distort the voltage waveform, and yet in several of them the true frequency stays essentially flat.
- That gap, an ugly voltage but a calm true frequency, is exactly what trips up the wrong estimator.

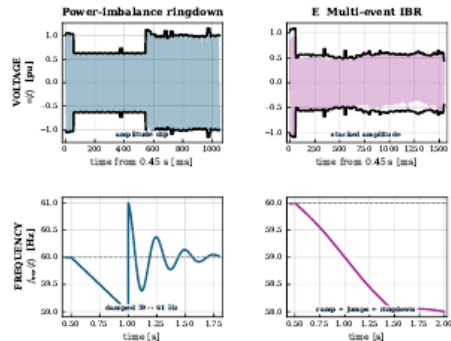
→ The last two composite cases are the genuinely nonstationary ones.

Key line: A distorted voltage does not mean the true frequency moved.

Composite IBR stresses (2 of 2): ringdown and multi-event

IBR STRESS SUITE

Composite IBR stresses (2 of 2): VOLTAGE $v(t)$ on top, true FREQUENCY $f(t)$ below



SG: Voltage on top, true frequency below: the estimator must recover through nonstationary segments, not just report low steady state error after the event

Say (verbatim):

- And here are the two hardest cases: a power-imbalance ringdown, and the stacked multi-event sequence, voltage on top and true frequency below.
- Now the frequency really does move, through damped oscillations and then a whole chain of events.
- The estimator has to track through these nonstationary segments, not just settle down afterward.

→ Before the methods, let me name the three mechanisms that actually break estimators.

Key line: Recovery through nonstationary segments is the real test.

Scenario families: distortion, leakage, and low SNR

Compact signal model

$$v(t) = A(t) \sin \theta(t) + v_h(t) + v_x(t) + n(t)$$

$$v_h(t) = \sum_{h=2}^H a_h \sin(h\theta + \phi_h), \quad v_x(t) = \sum_q b_q \sin(2\pi f_q t + \psi_q)$$

$$\text{THD} = \frac{\sqrt{\sum_{h \geq 2} V_h^2}}{V_1}, \quad \text{SNR}_{dB} = 20 \log_{10} \left(\frac{1/\sqrt{2}}{\sigma_n} \right)$$

Interpretation

The benchmark keeps the truth signal and the waveform perturbation separate. When $f_{\text{true}}(t)$ is fixed, estimator error measures artifact rejection, not a physical frequency event.

Family	What it tests
Harmonics	Integer multiples of 60 Hz; THD rejection with fixed true frequency.
Interharmonics	Non-integer tones; leakage and beating that can mimic dynamics.
OOB tones	Out-of-band interference; filter selectivity, aliasing margins, PLL leakage.
Noise / SNR	Controlled σ_n ; low-SNR crossings and covariance assumptions.

Why this slide matters

In these cases $f_{\text{true}}(t)$ can remain nominal. Any frequency excursion reported by the estimator is then a waveform-induced measurement artifact.

Scenario families: distortion, leakage, and low SNR

Say (verbatim):

- It helps to group the hard scenarios into three families, because each one breaks an estimator in a different way.
- Harmonic distortion attacks any method that assumes a pure sinusoid. Spectral leakage hurts the window-based methods when the frequency does not sit neatly inside a bin.
- And a low signal-to-noise ratio simply drowns the zero-crossing and timing-based methods.
- Keeping these three failure mechanisms separate is exactly what lets us explain, later, why a given method wins or collapses.

→ *With those mechanisms named, here are the estimators we put up against them.*

Key line: Distortion, leakage, and low SNR are three distinct failure mechanisms, not one.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL,
Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF,
LKF2.

Legacy baselines

TKEO and ZCD are retained as
comparison references outside the
recommended deployment set.

Window / spectral

IpDFT, TFT, ESPRIT, Prony,
MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different
latency and stress-sensitivity
positions. The comparison tracks
where each family wins, degrades,
and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Estimator families

Say (verbatim):

- The eighteen estimators sort naturally into families.
- Loop-based methods, the PLLs. Window and spectral methods. Model-based Kalman filters. Adaptive methods. And data-driven methods.
- We deliberately analyze by family, rather than method by method, because the family is what really explains the behavior we see, the strengths and the failure modes.

→ *And these are the numbers we score them on.*

Key line: Eighteen estimators, five families, one fair contract.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Metrics and decision variables

Say (verbatim):

- Here are the metrics. RMSE captures the average error. Peak error captures the worst moment.
- And then a trip-risk time, which counts how long the error stays above half a hertz, because that is where protection relays start to act.
- Plus settling time, and CPU cost. Five dimensions, kept separate on purpose.

→ Now let me show you what the data actually says.

Key line: Half a hertz is our protection-oriented risk proxy.

Core evidence (section)

Key line: The core evidence: five scenarios, from standard to composite.

Core evidence

From standard cases to composite IBR stress

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	0.006	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028
PLL	0.033	1.47	0.227
IpDFT	0.000	14.4	0.114
EKF	0.000	37.1	0.064

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Standard scenario results (build-up)

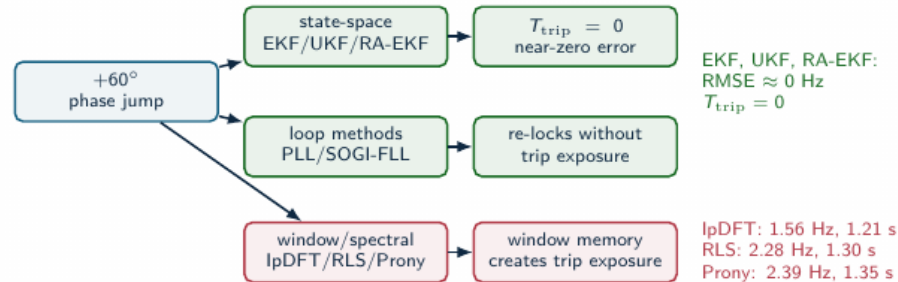
Say (verbatim):

- Let us start with the standard cases. On the step and the modulation, almost everything looks excellent, all of them below fifty millihertz.
- If we stopped here, we would conclude that the estimator choice barely matters.
- CLICK. But now watch the ramp column. The textbook EKF and the IpDFT intermittently blow up, to thirty-seven and fourteen hertz.
- And these are not edge methods; these are textbook, widely-used estimators.

→ You would expect a phase jump to be even worse, so let me show you that next.

Key line: Robustness, not point accuracy, is what separates the families.

A clean phase jump is a family separator



The clean jump does not expose RA-EKF's main advantage. It shows which estimator families carry memory through discontinuities.

Scenario D: phase jump separates families

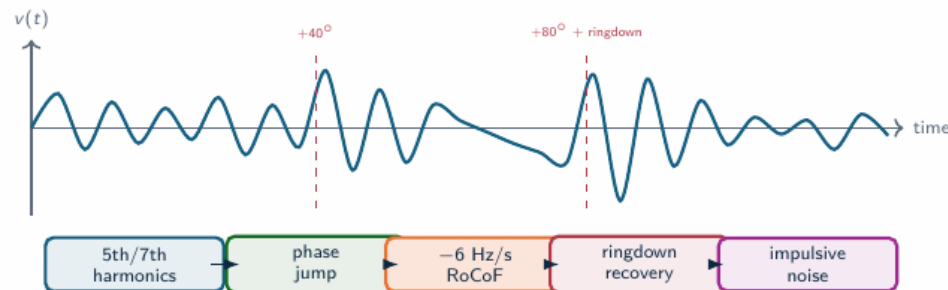
Say (verbatim):

- A clean sixty-degree phase jump cleanly separates the families.
- The state-space filters and the PLLs absorb it gracefully, with effectively zero trip exposure.
- The windowed and spectral methods, on the other hand, carry over a full second of trip exposure, because their analysis window straddles the discontinuity.
- But here is the part that surprised us.

→ *Because a single, clean jump is not actually the hard problem.*

Key line: The phase jump does not break the EKF; divergence on the ramp is the real danger.

Composite IBR sequence: why it is hardest



Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

Composite IBR sequence: hardest case

Say (verbatim):

- Scenario E stacks everything together: harmonics, two phase jumps, a rate-of-change segment, a ringdown, and impulsive noise.
- So spectral leakage, loop re-locking, and state-model mismatch all hit the estimator at the very same instant.
- There is nowhere for a method to hide.

→ *And this is exactly where the leaderboard falls apart.*

Key line: This is the hardest, most realistic case in the whole suite.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μ s]
UKF	1.09	0.78	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	0.78	11
SOGI-FLL	1.14	0.84	9
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	15 241
ZCD	426	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to ≈ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

Source: full_nc_benchmark, $n = 5$: IBR_Multi_Event

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000 $\times$ costlier; RA-EKF sits mid-pack but never diverges.

Scenario E: the leaderboard breaks

Say (verbatim):

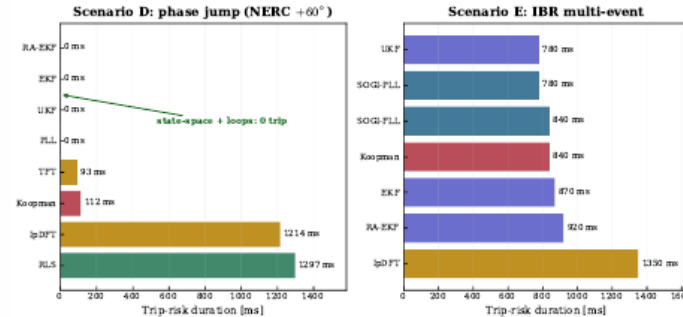
- Under that stacked sequence, every usable method collapses to roughly one-point-one hertz of error, and ZCD diverges all the way to four hundred hertz.
- There is no accurate winner here; high accuracy is simply not on the table.
- And worse, the rankings by RMSE, by trip-risk, and by CPU now disagree with one another. The method with the best RMSE is not the cheapest, and not the safest.

→ *Let me make the trip-risk side of that visual.*

Key line: Under stacked stress, there is simply no accurate winner.

Trip-risk evidence: core stress cases

Trip-risk is an event metric, not an average-error metric



Result reading

- Scenario D: state-space filters and PLL keep $T_{\text{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: every method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

Slide 31 / 67 ~23s cum 14:31

Trip-risk evidence (figure)

Say (verbatim):

- This is the trip-risk picture. In Scenario D, the state-space filters hold zero exposure, while the windowed methods sail past one-point-two seconds.
- But in Scenario E, look, everyone carries exposure. Nobody is safe.
- And notice the ranking here is completely different from the RMSE ranking.

→ *And you can actually watch the divergence happen, sample by sample.*

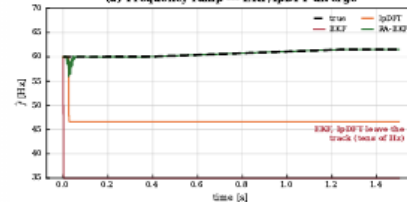
Key line: Trip-risk is a completely different ranking from RMSE.

The failures are visible in the trace, not only in the table

RAMP DIVERGENCE

Event responses (full_mc_benchmark): intermit

(a) Frequency ramp --- EKF/IpDFT diverge

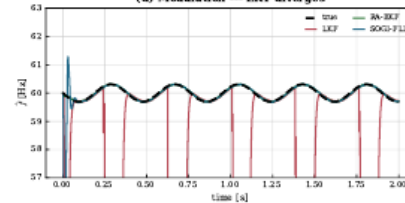


Failure signature

EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

MODULATION DIVERGENCE

(d) Modulation --- LKF diverges



Failure signature

LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.

Event responses (figure)

Say (verbatim):

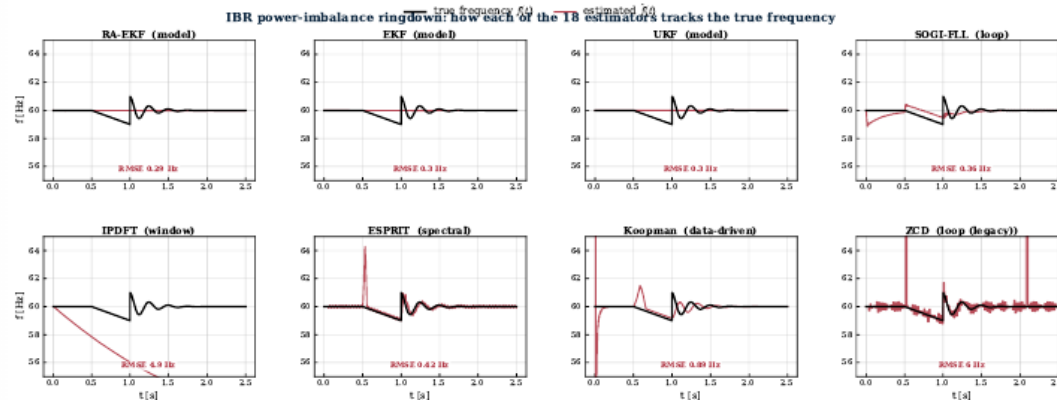
- Here it is, directly. The true frequency is the black line, and each colored line is an estimator trying to follow it.
- You can literally watch EKF and IpDFT run away on the ramp, climbing to tens of hertz, while RA-EKF just stays locked on the truth.
- This is what intermittent divergence looks like in practice.

→ Let me zoom into one hard event to see the tracking up close.

Key line: This is the intermittent divergence, seen with your own eyes.

How estimators track an IBR ringdown

FULL MONTE-CARLO



sg: Black is the true frequency, red is each estimator on the power-imbalance ringdown. Robust methods (RA-EKF, SOGI-FLL) follow the damped 50–61 Hz

How estimators track an IBR ringdown

Say (verbatim):

- Here is a closer look at one hard event, the power-imbalance ringdown, across eight representative estimators.
- Black is the true frequency, the damped swing between roughly fifty-nine and sixty-one hertz; red is each estimator's own output.
- The robust methods like RA-EKF and SOGI-FLL follow the swing cleanly, while fragile ones like ZCD overshoot or simply run away.

→ So what does all of this mean when you actually have to choose a method?

Key line: Robust methods track the damped swing; fragile ones overshoot or diverge.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0 ; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, lpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; lpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000 $\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 $\mu\text{s/sample}$; Koopman $\sim 15,000$	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Result comparison: winners by question

Say (verbatim):

- The honest summary is that the winner depends on the question you ask.
- Ask for clean steady-state accuracy, and the state-space methods and SOGI-FLL win.
- Ask for ramp robustness, and it is RA-EKF. Ask about the multi-event case, and there is no winner at all. Ask about feasibility, and the expensive methods simply drop out of contention.

→ *And that last point, feasibility, brings us straight to cost.*

Key line: Different question, different winner, every single time.

Computational feasibility

Method	Time/sample	Family	Target
SOGI-FLL	0.0348 μ s	Loop	MCU
SOGI-PLL	0.0867 μ s	Loop	MCU
EKF	0.813 μ s	Model	MCU/DSP
RA-EKF	1.61 μs	Model	DSP
TFT / IpDFT	3–4 μ s	Window	DSP/FPGA
Koopman	390 μ s	Data-driven	CPU/GPU
ESPRIT	2,838 μ s	Window	CPU

Source: SGSMA benchmark matrix, Sec. IV-D and Fig. 2(g)-(h)

Deployment interpretation

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

Key comparison

The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

Computational feasibility

Say (verbatim):

- Cost matters, because these run on relays and embedded hardware.
- The loop and state-space methods run in nine to seventeen microseconds per sample. That is comfortably embedded-feasible.
- The spectral and data-driven methods, by contrast, are a hundred to a thousand times slower. For a real-time protection device, that is often a deal-breaker, regardless of accuracy.

→ *Put cost and risk on the same picture, and it becomes even clearer.*

Key line: Cost alone already rules several methods out of a relay.

Deployment limits (figure)

Say (verbatim):



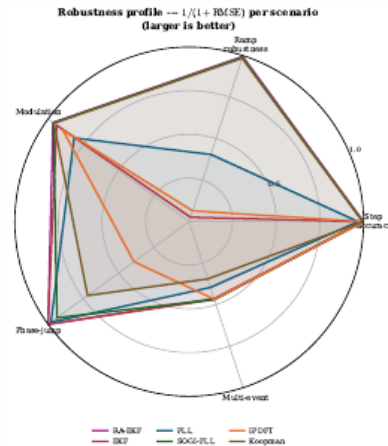
- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- In the OpenFreqBench expansion, state-space + loop methods cluster at low cost ($\leq 17 \mu\text{s}$) and low trip.
- Spectral/data-driven methods trade lower error for much higher CPU in this artifact.
- RA-EKF stays low-cost with no ramp divergence in this run.

Source: OpenFreqBench expansion: artifacts/full_nc_benchmark, $n = 5$

- Deployment is really the combination of all of it: accuracy, trip exposure, cost, and divergence-robustness, together.
 - And the encouraging thing in this picture is that the low-cost cluster, the state-space and loop methods, is also the low-trip cluster.
 - The cheap methods are often also the safe ones.
- *We can capture each method's overall balance in a single shape.*

Key line: The practical winners sit together, at low cost and low trip.

Balance profile



Radar profile

- Each axis is one scenario; larger = more robust ($1/(1 + RMSE)$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

Slide 37 / 67 ~23s cum 17:13

Balance profile (radar)

Say (verbatim):

- On this radar, each axis is one scenario, and a larger area means more robust.
- RA-EKF and SOGI-FLL stay robust on every single axis. EKF and IpDFT, in contrast, collapse on the ramp axis.
- The shape tells you the whole story at a glance.

→ Now, this next slide is the most important one about compliance testing.

Key line: No method is strong on every axis.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). P pass | F fails a declared RMSE/ T_{trip} /settling threshold | F* borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

Compliance heatmap (build-up)

Say (verbatim):

- This heatmap is the heart of the argument. Look at the standard columns first, step, ramp, and modulation. Almost every method passes.
- On a standard compliance test, these all look qualified.
- CLICK. But now the event columns appear, phase jump and multi-event, and watch: the same methods that passed now fail.
- Passing the standard tests told us almost nothing about event readiness.

→ *Let me distill all of this into our empirical claims.*

Key line: Passing the standard tests does not predict event readiness.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode.
On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{trip} = 0$.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Source: OpenFreqBench expansion: artifacts/full_nc_benchmark, $n = 5$

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
In the OpenFreqBench multi-event run, UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is $1000\times$ costlier.

Deployability

Feasibility changes the accuracy ranking.
In that same expansion, RA-EKF runs at $\sim 15 \mu\text{s/sample}$; selected spectral and data-driven methods move far outside that cost range.

Empirical claims

Say (verbatim):

- Four empirical claims come out of the core study.
- First, divergence, not phase jumps, is the dominant failure mode.
- Second, simple pass-fail testing leaves event readiness unresolved. Third, the three metrics, accuracy, trip-risk, and cost, pick genuinely different winners. And fourth, feasibility changes the ranking again.

→ Now the natural question is: does this pattern survive when we scale the study up?

Key line: Four claims, one theme: always report the conditions.

Expanded and ATLAS (section)

Expanded & ATLAS

Does the pattern scale across families and severity?

Key line: Does the pattern hold across families and severity?
That is ATLAS.

Expanded stress analysis

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Expanded stress analysis

Say (verbatim):

- To test that, we scale up to a full Monte-Carlo benchmark: eighteen estimators, thirty-three scenarios, five seeds each.
- And then we add the ATLAS severity sweeps on top, which push each disturbance from mild all the way to extreme.

→ *The first question is whether the family winners stay fixed as we scale.*

Key line: We scale the test, we do not just repeat it.



SGSMA benchmark: family winners

Family	Count	RMSE leader	RMSE mean	Trip-risk leader
IEEE ramps	9	SOGI-PLL	19.4 ± 6.4 mHz	0 s (multiple)
Magnitude steps	7	EKF	0.388 ± 0.143 mHz	0 s (multiple)
Modulation	3	RA-EKF	17.8 ± 21.4 mHz	0 s (multiple)
Phase jumps	3	SOGI-FLL	174 ± 67.5 mHz	ZCD: 0.0194 s
OOB interference	1	EKF	13.6 ± 11.1 mHz	EKF / RA-EKF: 0 s
IBR harmonics	3	EKF	70.5 ± 27.3 mHz	TFT / Prony: 0.00947 s
IBR ringdown	5	UKF	313 ± 23.2 mHz	ESPRIT: 0.0989 s
IBR multi-event	1	PLL	173.5 ± 66.7 mHz	lpDFT: 0.0439 s

The paper result is conditional: SOGI-PLL leads ramps, EKF leads magnitude steps/OOB/harmonics, RA-EKF leads modulation, UKF leads ringdown, and PLL leads the composite IBR case. Trip-risk often points elsewhere.

Source: SGSMA benchmark matrix, Table II

SGSMA benchmark: family winners

Say (verbatim):

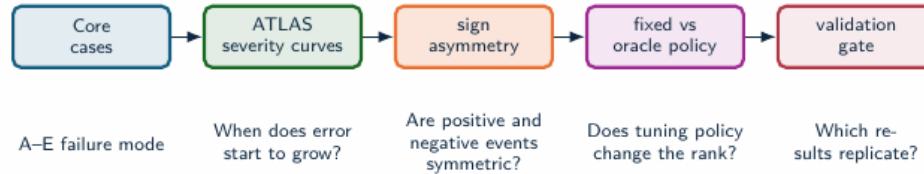
- And they do not stay fixed. The winner rotates from family to family.
- One method leads on the step, a different one on the ramp, another on harmonics, another on the multi-event case.
- There is no method that quietly wins everywhere when you are not looking.

→ *ATLAS lets us understand why, by sweeping the severity continuously.*

Key line: The leaderboard is regime-dependent, never absolute.

ATLAS severity analysis

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)



ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md

ATLAS severity analysis

Say (verbatim):

- ATLAS sweeps each stress by severity and, importantly, by sign, under a fixed policy, at thirty Monte-Carlo runs per point.
- And I want to be clear about evidence quality here: this is now paper-grade across all nine of the required sweeps, not a quick pilot.

→ *And the headline result from those sweeps is genuinely striking.*

Key line: This is paper-grade severity evidence, not a pilot.

ATLAS result: metric choice changes the answer

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Metric objective	Unique metric winner	Different from RMSE
Trip-risk exposure	26 / 113	5 / 26
RoCoF max error	112 / 113	53 / 112
RoCoF RMS error	112 / 113	48 / 112

Tie handling

Trip-risk has tied zero-exposure minima in 87/113 levels. It is an exposure screen first, not a single-winner leaderboard.

Why this matters

For protection-adjacent use, low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Operational implication

RoCoF metrics change the preferred estimator in nearly half the severity map; trip-risk removes estimators with long exposure.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv

ATLAS: metric choice changes the answer

Say (verbatim):

- Across a hundred and thirteen severity levels, the preferred estimator changes the moment you change the objective.
- In eighty-three of those levels, optimizing for RMSE and optimizing for trip-risk point you to two different methods.
- So the deliverable from ATLAS is not a ranking. It is a selection map: tell me your event and your metric, and it tells you the method.

→ *ATLAS also tells us something equally useful: where each method gives up.*

Key line: The output is a selection map, not a leaderboard.



ATLAS result: where estimators stop being valid

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

ATLAS: where estimators stop being valid

Say (verbatim):

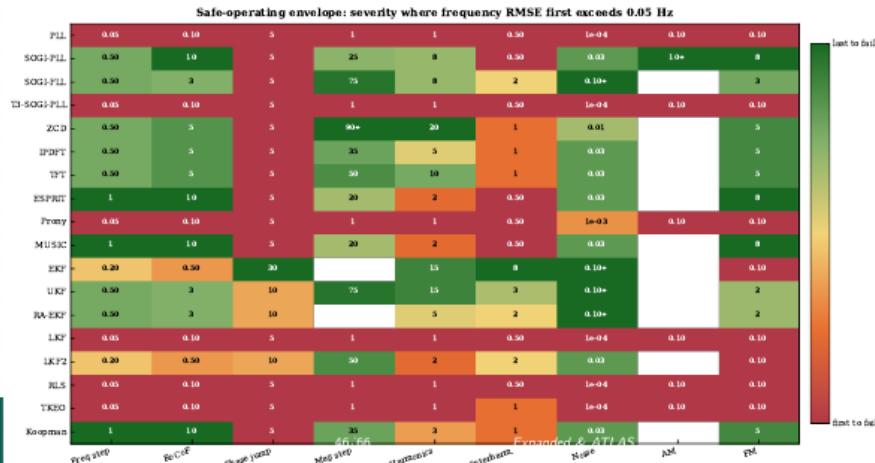
- For every method, ATLAS pinpoints the exact severity at which it crosses over from working to failing.
- That boundary, the point of no return, is different for every family, and knowing it is what lets you deploy a method safely.

→ And we can turn that into a single deployable map.

Key line: That validity boundary is different for every family.

Safe-operating envelope: where each estimator stays usable

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)



Safe-operating envelope

Say (verbatim):

- This is, for an engineer, the most useful single figure in the talk. Each cell is the severity at which a given estimator first exceeds the half-tenth-of-a-hertz error guide, expressed in that stress family's own units.
- Greener means the method tolerates more severe stress before failing; a plus sign means it never crossed the guide in the tested range.
- Read a row to see one estimator's safe envelope across all stresses; read a column to see who survives a given stress the longest. This is a map of operating limits, not a single ranking.

→ Now a few of the most surprising individual sweeps.

Key line: The deliverable is a map of safe operating limits per estimator, per stress.

AM, FM, and interharmonics split the story

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	SOGI-FLL, TFT, and Koopman split the low-rate points; ESPRIT leads from 1–10 Hz.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF leads 0.5–15%; PLL leads at 20%.	EKF crosses the 0.05 Hz guide between 5% and 8%; most families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one “distortion” label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

AM, FM and interharmonics

Say (verbatim):

- Even within modulation, the story splits in two.
- Amplitude modulation is relatively easy, and there the model-based methods win.
- But frequency modulation and interharmonics are much harder, and there the spectral methods take over. Same broad category, opposite winners.

→ *Now the handful of findings I most want you to walk away with.*

Key line: Even inside modulation, the winner switches.

When the textbook filter diverges

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

When the textbook filter diverges

Say (verbatim):

- This is the divergence mechanism up close. The textbook EKF tracks most seeds perfectly well, and then, on certain seeds, it intermittently runs away to tens of hertz.
 - And here is the unsettling part: it is not driven by severity. The worst seed in our set was actually the one with the gentlest ramp.
 - So you cannot predict it from how hard the event looks.
- *The next finding is about something we usually ignore: direction.*

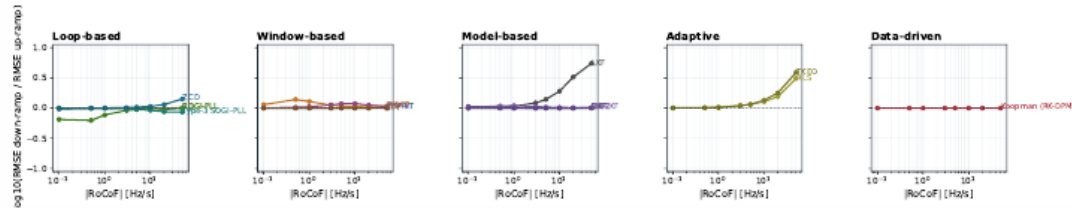
Key line: The robustified RA-EKF removes the divergence on every single seed.

RoCoF sign asymmetry: down-ramp sensitivity

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

RoCoF sign asymmetry by estimator family

Zero is sign-symmetric. Positive values mean the frequency-decline case is harder than the frequency-rise case.



Directional bias (RMSE₋/RMSE₊ at ± 50 Hz/s)

LKF is $5.5\times$ worse on $-RoCoF$; TKEO $3.9\times$, RLS $3.1\times$, SOGI-FLL $1.6\times$. EKF, UKF, and RA-EKF stay near 1.0.

Source: atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv ($n = 30$, both signs)

Why it matters

For these estimators, a $+RoCoF$ -only test can certify behavior that does not hold during frequency decline. Both signs must be swept.

Slide 49 / 67 ~25s cum 22:12

CORE

RoCoF has a sign

Say (verbatim):

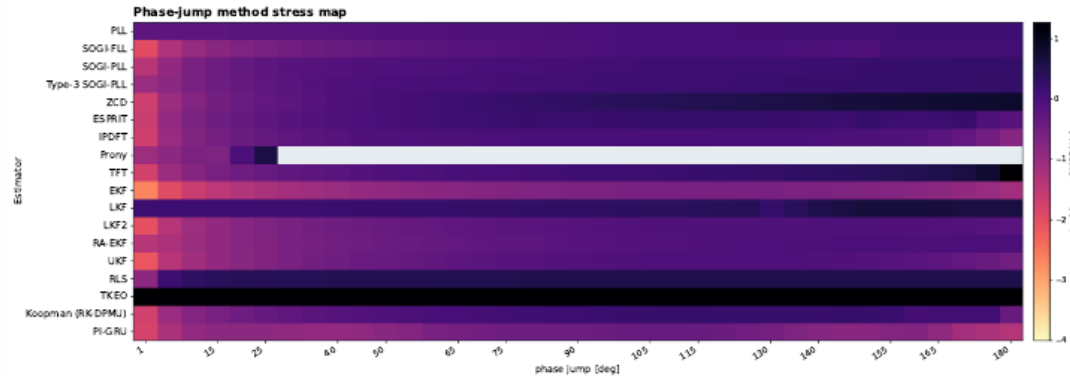
- Rate-of-change of frequency has a sign, and it turns out that down-ramps are harder than up-ramps.
 - The linear Kalman filter is five-and-a-half times worse when the frequency is falling. TKEO is almost four times worse.
 - This matters enormously, because a falling frequency is the protection-critical direction, the one that sheds load.
- And severity can surprise us in the opposite direction too.

Key line: A rising-only test can certify a method that fails when frequency drops.

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Most estimators peak before 180° ; TFT is the exception, with a late cliff near phase reversal.



Guide used in slide discussion: 0.05 Hz RMSE. Gray cells denote unavailable finite values.

Phase-jump severity is non-monotone

Say (verbatim):

- Phase-jump severity is non-monotone, which is genuinely counterintuitive. The worst case is not the largest jump.
- Most methods actually peak in error around ninety to a hundred and twenty degrees, and then recover toward a hundred and eighty, because a half-cycle jump partly cancels itself out.
- So if you test only at one nominal angle, you can completely miss the real worst case.

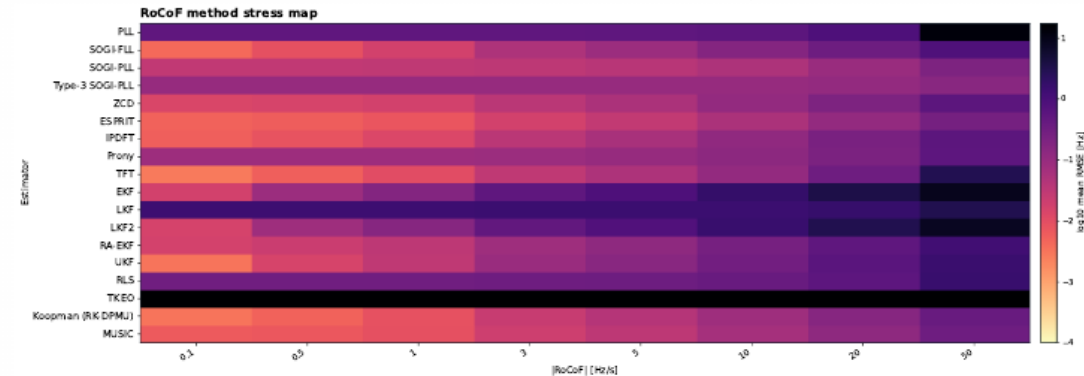
→ Severity also reshuffles the winners across the whole sweep.

Key line: Bigger is not always worse, so you have to sweep the range.

No estimator owns the whole stress space

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

RoCoF winners move by regime: Koopman at low stress, ESPRIT through the standard/IBR band, SOGI-PLL at severe RoCoF.



Guide used in slide discussion: 0.05 Hz RMSE. Gray cells denote unavailable finite values.

No universal winner (map)

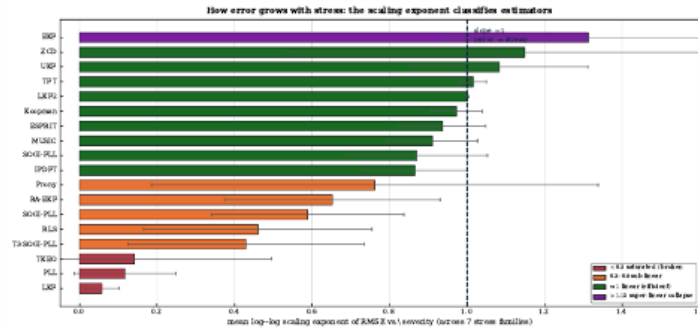
Say (verbatim):

- This map really captures the central message. The champion rotates by rate-of-change regime.
 - Koopman wins at low rates. ESPRIT wins in the middle. SOGI-PLL takes over once it gets severe.
 - No single column owns the whole map.
- And there is a deeper way to classify them, by how the error grows.

Key line: No universal winner; the deliverable is the map itself.

How error grows with stress: the scaling exponent

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)



Reading the exponent

The log-log slope of RMSE vs. severity is a physical signature:

- slope ≈ 1 : error grows linearly with stress — statistically efficient tracking (ESPRIT, MUSIC, TFT, UKF).
- slope < 0.3 : flat and bad — saturated or broken (TKEO, LKF).
- slope > 1.15 : super-linear collapse (EKF under noise).

Why it matters

The exponent, not the point error, predicts how a method behaves as an event worsens.

Source: atlas-papergrade-missing-v1/hypothesis_results.csv ($n = 30$, 7 stress families)

How error grows with stress: the scaling exponent

Say (verbatim):

- This is one of my favorite results, and it is more physics than benchmarking. If you take the log-log slope of error versus severity, you get a scaling exponent that classifies the estimators.
- A slope of about one, in green, means error grows linearly with stress — that is the Cramer-Rao-efficient regime, where the method is statistically as good as the noise allows. ESPRIT, MUSIC, TFT and UKF sit there.
- A slope near zero, in red, means flat and bad — saturated or broken, like TKEO and LKF. And a slope above one is super-linear collapse, which is where the textbook EKF lands under noise.
- The exponent, not the single error number, predicts how a method behaves as an event gets worse.

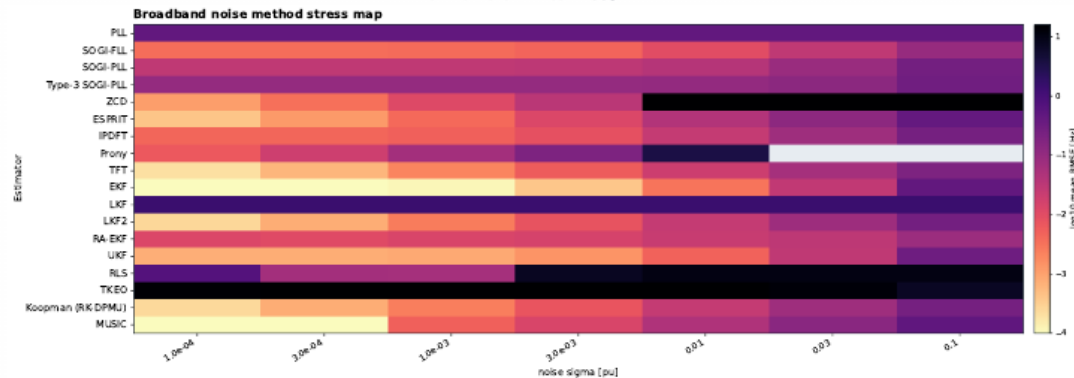
→ And one estimator shows this most dramatically, being both best and worst.

Key line: The scaling exponent classifies estimators by estimation theory, not by a single score.

ZCD is a narrow instrument, not a general estimator

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

ZCD is excellent at clean crossings and low THD. The same estimator collapses under broadband noise. It also collapses above the harmonic cliff near 20% THD.



Guides used in slide discussion: 0.05 Hz RMSSE. Gray cells demonstrate unavailable or finite values.

The ZCD paradox (build-up)

Say (verbatim):

- This is my favorite result, the ZCD paradox.
 - Below fifteen percent harmonic distortion, zero-crossing detection is the single most accurate method we tested. The best of all eighteen.
 - CLICK. But add broadband noise, and that very same method explodes to three thousand hertz.
 - CLICK. And push past twenty percent distortion, and it collapses again, to tens of hertz.
 - Best and worst, the same estimator, depending only on the stress.
- And the very same asymmetry shows up in voltage events.

Key line: Aggregate rankings would completely hide this.

Voltage sags and swells are not symmetric

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Estimator at $\pm 90\%$	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv; src/estimators/zcd.py

What to claim

ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Sags and swells are not symmetric

Say (verbatim):

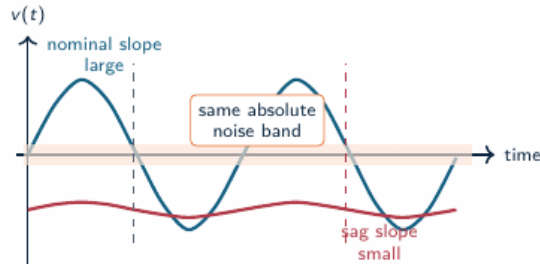
- Voltage magnitude is not symmetric either. A deep sag is far harder to handle than an equal swell.
- Zero-crossing detection goes from millihertz of error on a swell to hundreds of hertz on a deep sag.
- The direction of the voltage event, down versus up, changes the result by orders of magnitude.

→ And there is a clean, physical reason for that asymmetry.

Key line: Direction matters, so you have to sweep both signs.

Why a voltage sag breaks zero-crossing detection

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\left. \frac{dv}{dt} \right|_{t_0}} \quad \left. \frac{dv}{dt} \right|_0 \approx A\omega$$

When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

Slide 55 / 67 ~29s cum 25:26

FIG

Why a sag breaks zero-crossing

Say (verbatim):

- The mechanism is simple physics. The timing error of a zero-crossing scales with the noise divided by the amplitude times the frequency.
- A deep sag shrinks the amplitude, which flattens the slope of the waveform right at the crossing.
- And once the slope is flat, even small noise creates false crossings. It is not a coding bug; it is the geometry of the signal.

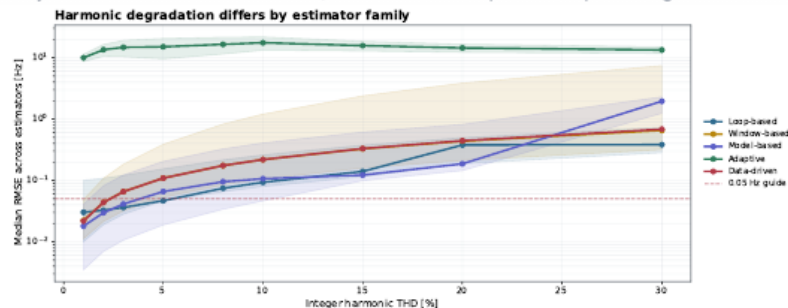
→ *Harmonics tell a similar, family-by-family story.*

Key line: This is physics, not a software bug.

Harmonic degradation differs by estimator family

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Family medians cross as THD increases; low-THD leadership does not predict high-THD survival.



Harmonic degradation by family

Say (verbatim):

- Harmonic degradation also depends strongly on the family.
- Loop methods like ZCD win at low distortion, but they collapse once you go above roughly fifteen percent.
- Window methods degrade much more gently, and they actually end up leading at extreme distortion. So low-distortion accuracy does not predict high-distortion survival.

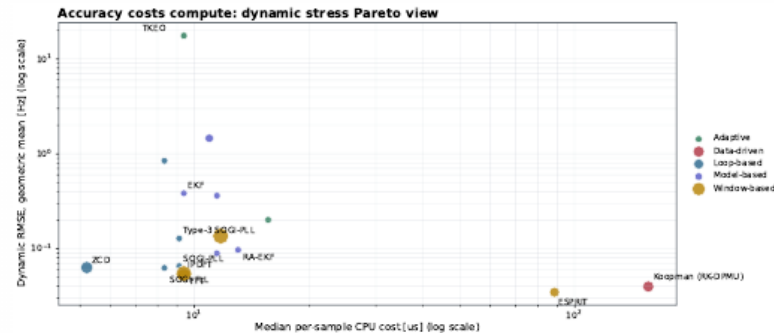
→ And cost is the final dimension to bring in.

Key line: Low-distortion accuracy does not predict high-distortion survival.

Accuracy costs 100–1000× more compute

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Deployable methods sit lower-left; spectral/data-driven accuracy moves right by 100–1000× CPU.



Slide 57 / 67 ~23s cum 26:13

CPU-accuracy Pareto

Say (verbatim):

- On the cost-accuracy Pareto front, the spectral and data-driven methods do buy you a little extra accuracy.
 - But they pay for it with a hundred to a thousand times more computation.
 - For relay-class, real-time timing, that trade is almost never worth it.
- So, with all of that on the table, what do we actually deploy?

Key line: For relay-class timing, that trade is usually not worth it.

Findings and deployment (section)

Key line: So what do we actually deploy?

Findings & deployment

What to deploy — and what comes next

Three findings

1. Metric-conditioned

The SGSMA matrix and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

2. Stress-bounded

AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy

The OpenFreqBench ramp seeds expose silent divergence; RA-EKF removes that failure mode in the tested run.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Three findings

Say (verbatim):

- Three findings sum up the work.
- First, there is a clear latency-versus-stress trade-off; the most accurate methods are often the slowest.
- Second, robustness beats point accuracy. RA-EKF, with its explicit rate-of-change state, stays stable exactly where the textbook EKF and IpDFT diverge.
- And third, compliance tests, on their own, are simply not enough for inverter-dominated events.

→ *Let me be honest about the limits of what we have shown.*

Key line: Robustness, not accuracy, is the deployment discriminator.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Validity limits and next steps

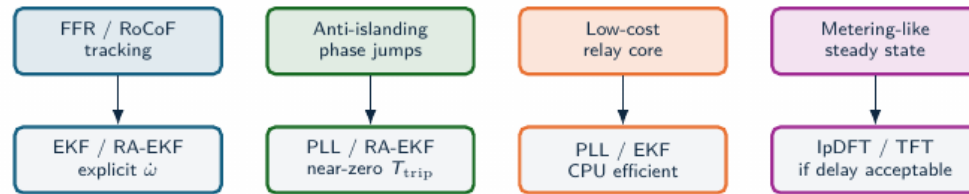
Say (verbatim):

- On validity, I want to be transparent. The core scenario means are at five seeds, and the ATLAS severity sweeps are paper-grade at thirty.
- The next step is journal-grade, at a hundred runs, and adding the subspace and machine-learning methods to the full sweep.
- We are deliberately explicit about what is diagnostic and what is paper-grade evidence.

→ *From there, the work becomes a concrete recommendation.*

Key line: We are explicit about what is diagnostic and what is paper-grade.

Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

Recommendation map

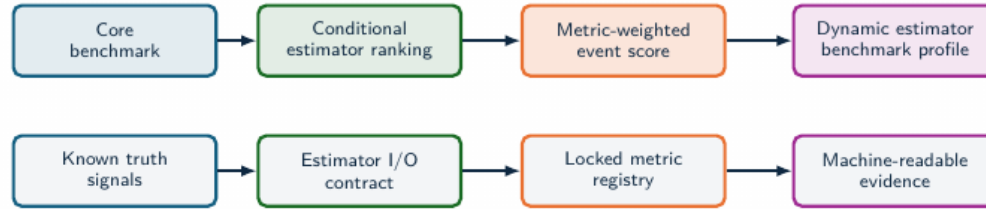
Say (verbatim):

- So here is our recommendation, and notice it is a map, not a single answer.
- For fast frequency response and anti-islanding, RA-EKF is the strongest, best-supported choice in our benchmark.
- PLL and EKF make solid low-cost baselines. And IpDFT or TFT are good where some observation delay is acceptable.

→ We then wrap that into a qualification idea.

Key line: We deliver a recommendation map, not a single ranking.

Qualification profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Qualification profile

Say (verbatim):

- A real qualification profile, in our view, should test transient recovery, trip exposure, compute feasibility, and standard-case RMSE, all together, not one at a time.
- Because, as we have seen, a method can pass any one of these and fail the others.

→ *And it should be reported in a standard, comparable form.*

Key line: Qualify on all dimensions at once, not one at a time.



Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

Standardization profile

Say (verbatim):

- So we propose reporting these dimensions explicitly and consistently, so that any two estimators can be compared on truly equal terms.
- That is what turns a one-off study into something the community can build on.

→ *And when you genuinely need just one number, here is how we do it.*

Key line: Make the comparison reproducible and fair.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in S} w_s \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

Score composition

RMSE	peak	trip	settle	cost
$\tilde{E}_s, \tilde{P}_s$	dynamic and peak error			
$\tilde{T}_s, \tilde{S}_s$	trip exposure and recovery			
$\tilde{C}_s, \tilde{L}_s$	compute and latency penalties			

Why a score is needed

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score makes the deployment objective explicit.

Protection profile

Raise γ, δ, λ when trip exposure, recovery, and latency dominate the risk model.

Monitoring profile

Raise α, β, η when reconstruction accuracy and compute budget dominate.

IBR Event Qualification Score

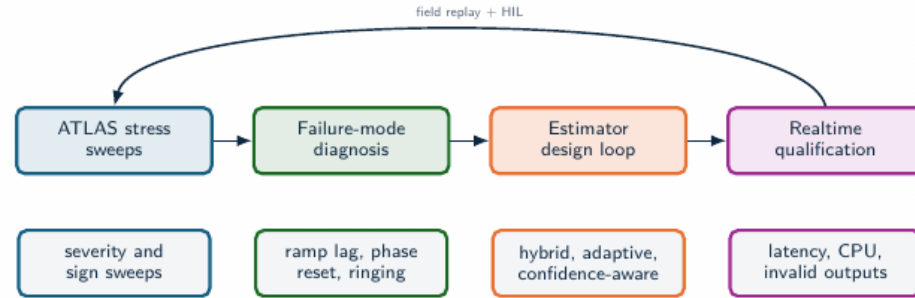
Say (verbatim):

- When a single number is unavoidable, for instance in a procurement spec, we define a composite, weighted, event-class qualification score.
- It rolls the dimensions into one figure, but it does so transparently, so you can always see what went into it.

→ Finally, let me point to where this work is heading.

Key line: One number when you need it, without hiding the dimensions.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Toward realtime IBR estimators

Say (verbatim):

- Looking ahead, there is a clear roadmap.
- Real-time estimators on hardware. Integrating recorded Simulink microgrid fault data. Adding modern and machine-learning methods. C++ acceleration for the heavy methods. And co-simulation with ANDES for closed-loop studies.

→ *And with that, let me close.*

Key line: This is where the benchmark goes next.



Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

Validation agenda and conclusion

Say (verbatim):

- So, to conclude. The single most important message is that there is no universal frequency estimator.
- The defensible choice always depends on the event you expect, the metric you care about, and the compute budget you have.
- And what OpenFreqBench provides is a reproducible, open way for the community to make that choice on evidence, rather than on habit.

Key line: Thank you very much for your attention. I would be very glad to take your questions.

Expert Q&A: anticipated hard questions



Likely question	Defensible answer
Is $n = 30$ enough for a statistical claim?	It clears our paper-grade gate with 95% CIs reported; we label evidence diagnostic vs. paper-grade and flag $n \geq 100$ as the replication layer. We claim mechanisms and rankings, not tight confidence bounds.
Why does the EKF diverge — is it a bug?	No. It is intermittent filter divergence: a seed-dependent onset/phase combination drives the covariance unstable. The worst seed had the gentlest ramp, so it is not severity-driven; the robustified RA-EKF removes it on every seed.
Scaling exponent ≈ 1 — coincidence?	It is the Cramer–Rao-efficient regime: error $\propto$ noise, so RMSE/σ is constant (verified). Slope-1 methods are statistically efficient; saturated methods sit near 0.
Isn't CPU time language-dependent?	Yes — it is a software-cost proxy from one Python run, not hardware latency. It separates feasible families ($\sim 10 \mu\text{s}$) from prohibitive ones ($\sim 10 \text{ms}$); absolute relay timing needs platform replication.
RA-EKF: overfitting to your own method?	It wins only ramp/multi-event robustness, not global RMSE; the leaderboard rotates by regime. Its advantage is an explicit $\hat{\omega}$ state and event gating, not tuning.
Why not real PMU data?	Known ground truth is the point: real streams lack a reference frequency, so error cannot be scored. Recorded Simulink/EMT fault data is on the roadmap.
ZCD 3000 Hz — is that physical?	Yes. Zero-cross timing error scales as $\sigma_R/(A\omega)$; once noise exceeds the slope at the crossing, false crossings appear, and the CI widens exactly at that bifurcation.

Expert Q&A: anticipated hard questions (backup)

Say (verbatim):

- (Backup slide — not presented; rehearse these answers for the question round.)
- Six hard questions to have ready: is $n=30$ enough (paper-grade gate + CIs, we claim mechanisms not tight bounds); is the EKF divergence a bug (no — intermittent filter divergence, seed-dependent, RA-EKF fixes it); is the slope-1 exponent a coincidence (no — Cramer-Rao efficiency); isn't CPU language-dependent (yes — it is a software-cost proxy, not hardware latency); are you overfitting RA-EKF (it wins only ramp/multi-event, not global RMSE); why not real PMU data (no ground truth to score against).
- Keep every answer to two sentences and always end by naming the evidence layer.

Key line: Answer in two sentences; always cite the evidence layer (diagnostic, paper-grade, or field-motivation).